Publications

Multi-Stage Knowledge Distillation with Layer Fusion-Based Deep Learning Approach for Skin Cancer Classification

Authors: Mahir Afser Pavel, Ramisa Asad, Md Ikramuzzaman, Murad Mustakim, and Riasat Khan

Abstract: Skin cancer is one of the most common types of cancer globally, caused by prolonged exposure to the sun's UV rays. Despite recent developments in research, early diagnosis, prevention, and treatment, skin cancer remains a significant health concern. This study proposes a multi-stage knowledge distillation-based deep learning technique with a layer fusion strategy to classify different types of skin lesion cells using the HAM10000 dataset. Augmentation and basic preprocessing steps have been proceeded on the HAM10000 dataset to enhance robustness. The applied multi-stage knowledge distillation incorporates intermediate features by measuring several loss values and coefficients to balance the corresponding losses. The proposed ViT and ConvNeXT-integrated teacher model leverages hybrid architectures derived from baseline models, combining convolutional feature extraction with transformer-based attention mechanisms. The distilled model, built using CNN and EfficientNet, achieved significant performance improvements over the baseline. The optimized model achieved an accuracy of 95.88%, F1 and AUC scores of 95.91% and 99.02%, respectively. Multi-stage knowledge distillation with intermediate exits and layer fusion improved model accuracy and performance metrics with the lowest training and inference times of 61 and 15 ms/step, respectively. Post-training quantization was applied to reduce the parameters and size of the distilled model. Multiple XAI techniques—Grad-CAM, Score-CAM, and LIME—have been explored to enhance the interpretability of the applied multi-stage knowledge distillation model. Domain expert evaluations favored Score-CAM as the most effective method for skin cancer detection due to its higher accuracy in highlighting key diagnostic features. The implementation codes with the quantized and distilled model are available in the following repository: https://github.com/codewith-pavel/Optimizations.

Keywords: Clinical decision-making, Explainable AI, Hybrid models, Knowledge distillation, Layer fusion, Post-training quantization, Skin cancer.

Journal: Scientific Reports, **Publisher**: nature, **DOI:** 10.1038/s41598-025-23403-2

Non-small cell lung cancer detection through knowledge distillation approach with teaching assistant

Authors: Mahir Afser Pavel, Rafiul Islam, Shoyeb Bin Babor, Riaz Mehadi, and Riasat Khan

Abstract: Non-small cell lung cancer (NSCLC) exhibits a comparatively slower rate of metastasis in contrast to small cell lung cancer, contributing to approximately 85% of the global patient population. In this work, leveraging CT scan images, we deploy a knowledge distillation technique within teaching assistant (TA) and student frameworks for NSCLC classification. We employed various deep learning models, CNN, VGG19, ResNet152v2, Swin, CCT, and ViT, and assigned roles as teacher, teaching assistant and student. Evaluation underscores exceptional model performance in performance metrics achieved via cost-sensitive learning and precise hyperparameter (alpha and temperature) fine-tuning, highlighting the model's efficiency in lung cancer tumor prediction and classification. The applied TA (ResNet152) and student (CNN) models achieved 90.99% and 94.53% test accuracies, respectively, with optimal hyperparameters (alpha = 0.7 and temperature = 7). The implementation of the TA framework improves the overall performance of the student model. After obtaining Shapley values, explainable AI is applied with a partition explainer to check each class's contribution, further enhancing the transparency of the implemented deep learning techniques. Finally, a web application designed to make it user-friendly and classify lung types in recently captured images. The execution of the three-stage knowledge distillation technique proved efficient with significantly reduced trainable parameters and training time applicable for memory-constrained edge devices.

Keywords: Non-small cell lung cancer, Adenocarcinoma of the lung, Distillation, Pulmonary imaging, Teachers, Squamous cell lung carcinoma, Lung and intrathoracic tumors, Web-based applications.

Journal: PloS one, **Publisher**: PLOS, **DOI:** 10.1371/journal.pone.0306441

Error-Tolerant Multimodal Vision-Language Models for Endodontic Triaging: A Cross-Sectional Study

Authors: Mahir Afser Pavel, Md Fahim Shahoriar Titu, Afifa Zain Apurba, Saif Ahmed, Shafin Rahman, James Dudley, and Taseef Hasan Farook

Abstract: To assess the performance of multimodal vision-language artificial intelligence models, optimised using quantisation-aware training, in triaging

endodontic treatment needs. The focus is on the ability to interpret endodontic radiographs while tolerating common image capture errors, including cone cutting, elongation, foreshortening, horizontal misalignment, over- and under-exposure and artefacts. In total 3600 dental radiographs were obtained across a 1-year period. Image augmentation techniques were applied to enhance model generalisability. Bootstrapped Language-Image Pretraining (BLIP), CLIP, Florence 2 and Paligemma multimodal models were fine-tuned using quantisation-aware training and evaluated using Bilingual Evaluation Understudy (BLEU), Recall-Oriented Understudy for Gisting Evaluation (ROUGE), Metric for Evaluation of Translation with Explicit ORdering (METEOR), Consensus-based Image Description Evaluation (CIDEr) and Loss Trends and Convergence. Quantisation-aware optimisation improved BLEU-4 by at least 17.3%, METEOR by at least 11.1%, ROUGE-L by at least 9.8% and CIDEr by at least 75.5% across all models. Quantisation reduced memory consumption by 87.5% while preserving diagnostic accuracy within a 0.5% error margin while correctly reproducing over 90% endodontic triage assessments made by practitioners. Multimodal AI demonstrates tolerance to imaging inconsistencies and is capable of accurately triaging endodontic cases with minimal computational demands, without compromising diagnostic performance.
Keywords: Endodontic diagnostics, Multimodal models, Quantization-aware training, Dental imaging, Orthopantomogram, Radiovisiography, Vision-language models.
Journal: International Journal of Dentistry, **Publisher**: Wiley, **DOI:** 10.1155/ijod/4148741

Fusion of Image Filtering and Knowledge-Distilled YOLO Models for Root Canal Failure Diagnosis

Authors: Afifa Zain Apurba, Md Fahim Shahoriar Titu, Mahir Afser Pavel, Intisar Tahmid Naheen, and Riasat Khan

Abstract: Root canal treatment involves the removal of inflamed pulp from infected teeth and sealing them to prevent re-entry of bacteria and infection. Early and accurate identification of root canal issues is crucial for improving treatment outcomes and minimizing complications. This study proposes a novel method for efficiently detecting failures in root canal treatments by incorporating image filtering techniques into deep learning models. A custom dataset has been collected from Square Hospital Ltd of Dhaka, Bangladesh, comprising 1,600 annotated, filtered radiographs. These images are processed using denoising algorithms (mean, median, Gaussian, contourlet transform, or Bayesian wavelet), non-local means, and BM3D. The YOLO models used for detection included YOLOv5, YOLOv7, and YOLOv8, with YOLOv5 achieving the highest accuracy of 90.6% on the filtered datasets. This work integrates an autodistillation technique with Grounded SAM as the base model for

bounding box generation, YOLOv8 as an intermediate model to bridge architectural differences, and YOLOv5 as the target model due to its lightweight design, fewer parameters, and superior performance on the filtered dataset. Knowledge distillation was employed to optimize performance on lightweight models, resulting in precision and recall values of 99.16% and 97.29%, respectively, for YOLOv5. The filtered datasets outperformed raw images, with the Bayesian wavelet and contourlet transform yielding the best results on YOLOv5. This pipeline demonstrates the potential to develop one of the most accurate and clinically applicable diagnostic tools for root canal failure detection by combining deep learning and knowledge distillation techniques with image filtering. This study demonstrates how combining deep learning with advanced image enhancement techniques can provide a highly accurate, clinically relevant tool for the early detection of root canal treatment failures, potentially aiding dentists in making faster and more consistent diagnoses.
Keywords: Root canal failure detection, filtered radiographs, YOLO models, knowledge distillation, image filtering techniques, deep learning in dentistry.
Journal: IEEE Access, **Publisher**: IEEE, **DOI:** 10.1109/ACCESS.2025.3560998

Real-Time Fire Detection: Integrating Lightweight Deep Learning Models on Drones with Edge Computing

Authors: Md Fahim Shahoriar Titu, Mahir Afser Pavel, Goh Kah Ong Michael, and Riasat Khan

Abstract: Fire accidents are life-threatening catastrophes leading to losses of life, financial damage, climate change, and ecological destruction. Promptly and efficiently detecting and extinguishing fires is essential to reduce the loss of lives and damage. This study uses drone, edge computing, and artificial intelligence (AI) techniques, presenting novel methods for real-time fire detection. This proposed work utilizes a comprehensive dataset of 7187 fire images and advanced deep learning models, e.g., Detection Transformer (DETR), Detectron2, You Only Look Once YOLOv8, and Autodistill-based knowledge distillation techniques to improve the model performance. The knowledge distillation approach has been implemented with the YOLOv8m (medium) as the teacher (base) model. The distilled (student) frameworks are developed employing the YOLOv8n (Nano) and DETR techniques. The YOLOv8n attains the best performance with 95.21% detection accuracy and 0.985 F1 score. A powerful hardware setup, including a Raspberry Pi 5 microcontroller, Pi camera module 3, and a DJI F450 custom-built drone, has been constructed. The distilled YOLOv8n model has been deployed in the proposed hardware setup for real-time fire identification. The YOLOv8n model achieves 89.23% accuracy and an approximate frame rate of 8 for the conducted live experiments. Integrating deep learning techniques with drone and edge devices demonstrates the proposed system's effectiveness and potential for practical applications in fire hazard mitigation.

Keywords: artificial intelligence; deep learning; edge computing; knowledge distillation; Raspberry Pi 5; real-time fire detection; you only look once.
Journal: Drones, **Publisher**: MDPI, **DOI:** 10.3390/drones8090483

Hybrid ViT-RetinaNet with Explainable Ensemble Learning for Fine-Grained Vehicle Damage Classification

Authors: Ananya Saha, Mahir Afser Pavel, Afifa Zain Apurba, Md Fahim Shahoriar Titu, and Riasat Khan

Abstract: Efficient and explainable vehicle damage inspection is essential due to the increasing complexity and volume of vehicular incidents. Traditional manual inspection approaches are not time-effective, prone to human error, and lead to inefficiencies in insurance claims and repair workflows. Existing deep learning methods, such as CNNs, often struggle with generalization, require large annotated datasets, and lack interpretability. This study presents a robust and interpretable deep learning framework for vehicle damage classification, integrating Vision Transformers (ViTs) and ensemble detection strategies. The proposed architecture employs a RetinaNet backbone with a ViT-enhanced detection head, implemented in PyTorch using the Detectron2 object detection technique. It is pretrained on COCO weights and fine-tuned through focal loss and aggressive augmentation techniques to improve generalization under real-world damage variability. The proposed system applies the Weighted Box Fusion (WBF) ensemble strategy to refine detection outputs from multiple models, offering improved spatial precision. To ensure interpretability and transparency, we adopt numerous explainability techniques—Grad-CAM, Grad-CAM++, and SHAP—offering semantic and visual insights into model decisions. A custom vehicle damage dataset with 4500 images has been built, consisting of approximately 60% curated images collected through targeted web scraping and crawling covering various damage types (such as bumper dents, panel scratches, and frontal impacts), along with 40% COCO dataset images to support model generalization. Comparative evaluations show that Hybrid ViT-RetinaNet achieves superior performance with an F1-score of 84.6%, mAP of 87.2%, and 22 FPS inference speed. In an ablation analysis, WBF, augmentation, transfer learning, and focal loss significantly improve performance, with focal loss increasing F1 by 6.3% for underrepresented classes and COCO pretraining boosting mAP by 8.7%. Additional architectural comparisons demonstrate that our full hybrid configuration not only maintains competitive accuracy but also achieves up to 150 FPS, making it well suited for real-time use cases. Robustness tests under challenging conditions, including real-world visual disturbances (smoke, fire, motion blur, varying lighting, and occlusions) and artificial noise (Gaussian; salt-and-pepper), confirm the model’s generalization ability. This work contributes a scalable, explainable, and high-performance solution for real-world vehicle damage diagnostics.

Keywords: vehicle damage classification; vision transformer; RetinaNet; Detectron2; explainable AI; ensemble modeling; weighted box fusion; insurance automation
Journal: vehicles, **Publisher**: MDPI, **DOI:** 10.3390/vehicles7030089

Under Review

S³-Net: Self-Stabilizing, Self-Regulating, and Self-Optimizing Neural Networks via Universal Stability Learning

Authors: Anonymous Authors

Abstract: Deep neural networks have achieved remarkable performance across a wide range of computer vision tasks; however, their training processes remain largely passive, with optimization strategies, learning schedules, and regularization parameters typically predetermined rather than dynamically adapting to the evolving stability of the learning process. This limitation can increase sensitivity to initialization, hyperparameter configurations, and stochastic optimization dynamics, resulting in inconsistent convergence and reduced training reliability. To address this challenge, this study presents S³-Net, a unified framework for Self-Stabilizing, Self-Regulating, and Self-Optimizing Neural Networks via Universal Stability Learning, which reformulates neural network training as a closed-loop optimization process. S³-Net integrates a stability-aware FRFormer backbone with a Dynamic Stability Module (DSM), Neural Stability Controller (NSC), Meta Optimizer (MO), Universal Stability Loss (USL), and Stability Distillation (SD). The DSM continuously characterizes feature, activation, prediction, and gradient dynamics, while the NSC uses the resulting stability state to adapt the training process online. The Meta Optimizer learns transferable optimization configurations, whereas USL explicitly incorporates stability into the training objective and SD transfers stability-aware representations across different model capacities. The framework is evaluated across 6 datasets and 8 representative architectures under multiple random seeds, hyperparameter perturbations, and training budgets. Compared with corresponding baseline training strategies, S³-Net improves average predictive performance by 3.27% while reducing the aggregate instability measure by 41.36%. Across independent training runs, the framework reduces performance variance by 46.82% and sensitivity to hyperparameter perturbations by 42.57%. The adaptive controller introduces an average computational overhead of only 8.73%, while Stability Distillation reduces the teacher–student stability gap by 34.68%. Comprehensive ablation studies demonstrate that stability monitoring, adaptive regulation, stability-aware optimization, and stability transfer provide complementary improvements, with the complete framework achieving the most consistent performance across the evaluated settings. Overall, these findings support

autonomous stability learning as a general approach for transforming conventional open-loop neural optimization into a more adaptive, self-regulating, and reliable learning process.
Keywords: Deep neural networks; Training stability; Adaptive optimization; Hyperparameter robustness; Meta-optimization; Feature dynamics; Stability learning; Universal stability loss; Closed-loop training.

ACP-KD: Adaptive Clinical Prototype Knowledge Distillation with Dynamic Feature Fusion and Confidence-Aware Multi-Objective Learning for Lightweight Skin Lesion Classification

Authors: Anonymous Authors
Abstract: Automated skin lesion classification has achieved substantial progress through deep learning; however, the computational complexity of high-capacity models remains a major barrier to deployment in resource-constrained clinical environments. Knowledge distillation offers an effective approach for transferring diagnostic knowledge from powerful teacher networks to lightweight student models, yet conventional methods largely rely on output-level supervision and may fail to capture the heterogeneous, sample-dependent, and reliability-sensitive nature of medical image representations. To address these limitations, this study proposes Adaptive Clinical Prototype Knowledge Distillation (ACP-KD), a multi-objective framework designed to improve the efficiency, robustness, and calibration of lightweight skin lesion classifiers. ACP-KD employs a high-capacity hybrid teacher and a lightweight student architecture with dynamic feature fusion, enabling the transfer of complementary diagnostic knowledge through multiple distillation objectives, including logit, attention, feature, prototype, concept, and confidence distillation. A difficulty-aware adaptive weighting mechanism dynamically adjusts the contribution of these knowledge sources according to individual sample characteristics and predictive reliability. Furthermore, prototype-guided representation learning promotes discriminative class-aware feature spaces, while counterfactual regularization enhances the robustness of learned representations. Confidence-aware supervision is incorporated to improve predictive calibration and reliability. Comprehensive experiments on the HAM10000 dataset, including ablation, calibration, computational efficiency, and cross-dataset evaluations, demonstrate the effectiveness of the proposed framework. ACP-KD achieves 95.27% accuracy, 95.10% macro-F1, and 0.992 AUC, while substantially reducing computational requirements compared with the high-capacity teacher model. The framework also achieves an Expected Calibration Error (ECE) of 0.014, indicating improved predictive reliability and calibration. Extensive ablation studies further confirm the complementary contributions of the individual knowledge sources and adaptive mechanisms. Overall, the findings demonstrate that ACP-KD provides an effective pathway for transferring diverse high-capacity diagnostic knowledge into efficient,

robust, and better-calibrated lightweight models, supporting their potential deployment in resource-constrained medical imaging applications.
Keywords: Skin lesion classification; Knowledge distillation; Medical image analysis; Dynamic feature fusion; Prototype learning; Confidence calibration; Model efficiency; HAM10000; Explainable and reliable AI.

Trustworthy Clinical AI for Diabetic Retinopathy Grading via Hybrid Foundation Models, Calibration-Aware Uncertainty Learning, and Multi-Criteria Clinical Referral

Authors: Anonymous Authors
Abstract: Diabetic retinopathy (DR) is a leading cause of preventable blindness worldwide, making trustworthy automated screening essential for early diagnosis and clinical intervention. Although recent deep learning approaches have achieved promising classification performance, their clinical adoption remains constrained by poor confidence calibration, unreliable uncertainty estimation, limited interpretability, and simplistic referral strategies. This study proposes TRUST-DR, a trustworthy clinical artificial intelligence framework for automated DR grading that integrates hybrid CNN–foundation model feature learning, calibration-aware uncertainty estimation, quantitative explainability, clinical trust estimation, and a learnable multi-criteria referral mechanism within a unified end-to-end architecture. A hybrid ResNet50–DINOv2 backbone captures complementary local lesion-level and global retinal representations, while predictive uncertainty, confidence calibration, image quality, and Grad-CAM-derived explainability are jointly incorporated to estimate a continuous Clinical Trust Score and guide intelligent referral decisions for clinically ambiguous cases. Experiments on the APTOS 2019 dataset demonstrate that TRUST-DR achieves an accuracy of 95.27%, a Quadratic Weighted Kappa (QWK) of 0.9824, an Expected Calibration Error (ECE) of 0.014, and a Brier Score of 0.059. These results demonstrate the potential of TRUST-DR to provide an accurate, well-calibrated, transparent, and clinically trustworthy decision-support framework for automated diabetic retinopathy screening.
Keywords: Diabetic retinopathy; Trustworthy AI; Medical image analysis; Hybrid foundation models; Calibration-aware learning; Uncertainty estimation; Clinical referral; APTOS 2019; Retinal image grading.

CLKD-MED: A novel cross-lingual knowledge distillation framework for multilingual clinical outcome prediction

Authors: Mahir Afser Pavel, Rafiul Islam, Mohammad Junayed Hasan, and M. R. C. Mahdy
Abstract: Current clinical outcome prediction systems are predominantly monolingual, limiting their applicability in linguistically diverse healthcare settings. This study introduces CLKD-MED, the first cross-lingual knowledge distillation framework explicitly designed for clinical outcome modeling, enabling robust deployment across low-resource languages without requiring extensive annotated clinical corpora. Our algorithm integrates prompt-conditioned neural translation with cross-lingual knowledge distillation to preserve medical semantics during transfer and improve translation fidelity. CLKD-MED leverages ensemble teacher models trained on English clinical texts to supervise compact student models in the target language, incorporating SHAP-based interpretability to ensure clinically transparent decision-making. Evaluated on the MIMIC-III clinical database with Bengali as the low-resource target, CLKD-MED achieves ROC-AUC scores of 0.81 for length of stay and 0.83 for mortality prediction. Compared to direct transfer learning, XLM-RoBERTa gains up to 0.067 AUC and 2–3× inference efficiency under cross-lingual distillation, establishing a new paradigm for scalable, equitable, and explainable clinical AI.
Keywords: Clinical Text Processing, Information Preservation, Multi-Granularity Analysis, Adaptive Weighting, Medical NLP

A Federated Multi-Component Framework for Brain Tumor Diagnosis with Generative Augmentation and Explainable Representation Learning

Authors: Mahir Afser Pavel, Nafiz Fahad, Md Tanzib Hosain, Md. Kishor Morol, Tze Hui Liew
Abstract: Brain tumors remain a major challenge in neuro-oncology due to diagnostic complexity, inter-observer variability, and increasing demand for reliable and privacy-aware computational analysis. Although deep learning has improved imaging-based diagnosis, practical deployment remains constrained by data decentralization, robustness of representations, class imbalance, and limited interpretability. This study presents a federated multi-component framework for brain tumor diagnosis that combines privacy-preserving federated optimization, generative augmentation, dual view representation learning, contrastive feature learning, and explainable artificial intelligence. Rather than introducing a new learning paradigm, the framework is designed to investigate how these complementary components interact to improve representation quality, robustness, and interpretability under decentralized training settings. Experimental evaluation on the Brain MRI ND-5 dataset demonstrated classification performance exceeding 96% across Accuracy and F1-score. Component-wise ablation studies and comparative analyses indicated that the integrated configuration consistently improved

performance relative to individual modules and baseline architectures while maintaining privacy-aware distributed training and interpretable decision support. The findings suggest that combining federated learning, generative augmentation, contrastive representation learning, and explainability can provide a practical and scalable approach for MRI-based brain tumor diagnosis. Future work will focus on broader multi-institutional validation, multimodal integration, efficient training strategies, and further assessment of clinical usability and generalization.
Keywords: Neuro-oncology, Federated Learning, Generative Adversarial Network, Dual Encoder, Contrastive Learning, Explainable AI

SLGRA-Net: Structure-Aware Latent Graph Reasoning for 3D Brain Tumor Segmentation

Authors: Mahir Afser Pavel, Nafiz Fahad, Md Tanzib Hosain, Md. Kishor Morol, Tze Hui Liew

Abstract: Segmentation of brain tumor subregions from multi-modal 3D MRI is necessary for clinical diagnosis and treatment planning. However, it continues to pose significant challenges due to severe class imbalance, ambiguous boundaries, and fragmented enhancing tumor (ET) regions. While convolutional and transformer-based models improve local and global feature representations, most methods operate in dense voxel or token spaces, resulting in high computational cost and a lack of explicit structural representation. This paper proposes SLGRA-Net, a Structure-aware Latent Graph Reasoning Network that enables global structural reasoning in a compact latent graph space. The framework aggregates 3D feature patches into graph nodes, allowing the efficient capture of long-range structural dependencies while preserving voxel-level precision. Graph-based message passing refines node features, which are adaptively fused into the decoder via gated unpooling. To address severe class imbalance and label uncertainty, particularly for ET, we introduce an auxiliary node-level supervision mechanism that is uncertainty-aware and employs class-balanced weighting. Extensive experiments on the BraTS benchmarks demonstrate that SLGRA-Net achieves strong performance (0.89 Dice), particularly for the clinically critical ET region, while significantly reducing computational overhead compared to transformer-based methods. Comprehensive ablation studies validate each architectural component, and external validation confirms the framework’s robust and consistent performance across diverse datasets.
Keywords: Brain tumor segmentation, latent graph reasoning, structure-aware learning, multi-modal MRI, and uncertainty-aware modeling.